Configuration stm32-foc
STM32CubeMX 6.18.1
Date 08/27/2026
MCU STM32F303RETx

PERIPHERALS	MODES	FUNCTIONS	PINS
ADC1	IN1 Single-ended	ADC1_IN1	PA0
ADC1	IN2 Single-ended	ADC1_IN2	PA1
ADC1	IN6 Single-ended	ADC1_IN6	PC0
ADC1	IN7 Single-ended	ADC1_IN7	PC1
ADC2	IN1 Single-ended	ADC2_IN1	PA4
ADC2	IN5 Single-ended	ADC2_IN5	PC4
ADC2	IN8 Single-ended	ADC2_IN8	PC2
ADC2	IN9 Single-ended	ADC2_IN9	PC3
ADC2	IN11 Single-ended	ADC2_IN11	PC5
RCC	BYPASS Clock Source	RCC_OSC_IN	PF0-OSC_IN
RCC	Crystal/Ceramic Resonator	RCC_OSC32_IN	PC14-OSC32_IN
RCC	Crystal/Ceramic Resonator	RCC_OSC32_OUT	PC15-OSC32_OUT
SYS	Serial Wire	SYS_JTCK-SWCLK	PA14
SYS	Serial Wire	SYS_JTMS-SWDIO	PA13
SYS	TIM17	SYS_VS_tim17	VP_SYS_VS_tim17
TIM1	Internal Clock	TIM1_VS_ClockSourceINT	VP_TIM1_VS_ClockSourceINT
TIM1	PWM Generation CH1 CH1N	TIM1_CH1	PA8
TIM1	PWM Generation CH1 CH1N	TIM1_CH1N	PA7
TIM1	PWM Generation CH2 CH2N	TIM1_CH2	PA9
TIM1	PWM Generation CH2 CH2N	TIM1_CH2N	PB0
TIM1	PWM Generation CH3 CH3N	TIM1_CH3	PA10
TIM1	PWM Generation CH3 CH3N	TIM1_CH3N	PB1
TIM1	PWM Generation No Output	TIM1_VS_no_output4	VP_TIM1_VS_no_output4
TIM1	Activate-Break-Input	TIM1_BKIN	PA6
USART2	Asynchronous	USART2_RX	PA3
USART2	Asynchronous	USART2_TX	PA2

Pin_Nb	PINs	FUNCTIONs	LABELs
2	PC13	GPIO_EXTI13	B1 [Blue PushButton]
3	PC14-OSC32_IN	RCC_OSC32_IN	
4	PC15-OSC32_OUT	RCC_OSC32_OUT	
5	PF0-OSC_IN	RCC_OSC_IN	
6	PF1-OSC_OUT*	RCC_OSC_OUT	
8	PC0	ADC1_IN6	CURR_W
9	PC1	ADC1_IN7	CURR_V
10	PC2	ADC2_IN8	TEMP
11	PC3	ADC2_IN9	BEMF_U
14	PA0	ADC1_IN1	CURR_U
15	PA1	ADC1_IN2	V_BUS
16	PA2	USART2_TX	USART_TX
17	PA3	USART2_RX	USART_RX
20	PA4	ADC2_IN1	POTEN
21	PA5	GPIO_Output	LD2 [Green Led]
22	PA6	TIM1_BKIN	
23	PA7	TIM1_CH1N	
24	PC4	ADC2_IN5	BEMF_V

25	PC5	ADC2_IN11	BEMF_W
26	PB0	TIM1_CH2N	
27	PB1	TIM1_CH3N	
28	PB2	GPIO_Output	LED_RED
40	PC9	GPIO_Output	GPIO_BEMF
41	PA8	TIM1_CH1	
42	PA9	TIM1_CH2	
43	PA10	TIM1_CH3	
46	PA13	SYS_JTMS-SWDIO	TMS
49	PA14	SYS_JTCK-SWCLK	TCK
55	PB3*	SYS_JTDO-TRACESWO	SWO

PERIPHERALS	MODES	FUNCTIONS	PINS
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ADC1	IN2 Single-ended	ADC1_IN2	PA1
ADC1	IN6 Single-ended	ADC1_IN6	PC0
ADC1	IN7 Single-ended	ADC1_IN7	PC1
ADC2	IN1 Single-ended	ADC2_IN1	PA4
ADC2	IN5 Single-ended	ADC2_IN5	PC4
ADC2	IN8 Single-ended	ADC2_IN8	PC2
ADC2	IN9 Single-ended	ADC2_IN9	PC3
ADC2	IN11 Single-ended	ADC2_IN11	PC5
RCC	BYPASS Clock Source	RCC_OSC_IN	PF0-OSC_IN
RCC	Crystal/Ceramic Resonator	RCC_OSC32_IN	PC14-OSC32_IN
RCC	Crystal/Ceramic Resonator	RCC_OSC32_OUT	PC15-OSC32_OUT
SYS	Serial Wire	SYS_JTCK-SWCLK	PA14
SYS	Serial Wire	SYS_JTMS-SWDIO	PA13
SYS	TIM17	SYS_VS_tim17	VP_SYS_VS_tim17
TIM1	Internal Clock	TIM1_VS_ClockSourceINT	VP_TIM1_VS_ClockSourceINT
TIM1	PWM Generation CH1 CH1N	TIM1_CH1	PA8
TIM1	PWM Generation CH1 CH1N	TIM1_CH1N	PA7
TIM1	PWM Generation CH2 CH2N	TIM1_CH2	PA9
TIM1	PWM Generation CH2 CH2N	TIM1_CH2N	PB0
TIM1	PWM Generation CH3 CH3N	TIM1_CH3	PA10
TIM1	PWM Generation CH3 CH3N	TIM1_CH3N	PB1
TIM1	PWM Generation No Output	TIM1_VS_no_output4	VP_TIM1_VS_no_output4
TIM1	Activate-Break-Input	TIM1_BKIN	PA6
USART2	Asynchronous	USART2_RX	PA3
USART2	Asynchronous	USART2_TX	PA2

Pin Nb	PINs	FUNCTIONs	LABELs
2	PC13	GPIO_EXTI13	B1 [Blue PushButton]
3	PC14-OSC32_IN	RCC_OSC32_IN	
4	PC15-OSC32_OUT	RCC_OSC32_OUT	
5	PF0-OSC_IN	RCC_OSC_IN	
6	PF1-OSC_OUT*	RCC_OSC_OUT	
8	PC0	ADC1_IN6	CURR_W
9	PC1	ADC1_IN7	CURR_V
10	PC2	ADC2_IN8	TEMP
11	PC3	ADC2_IN9	BEMF_U
14	PA0	ADC1_IN1	CURR_U

15	PA1	ADC1_IN2	V_BUS
16	PA2	USART2_TX	USART_TX
17	PA3	USART2_RX	USART_RX
20	PA4	ADC2_IN1	POTEN
21	PA5	GPIO_Output	LD2 [Green Led]
22	PA6	TIM1_BKIN	
23	PA7	TIM1_CH1N	
24	PC4	ADC2_IN5	BEMF_V
25	PC5	ADC2_IN11	BEMF_W
26	PB0	TIM1_CH2N	
27	PB1	TIM1_CH3N	
28	PB2	GPIO_Output	LED_RED
40	PC9	GPIO_Output	GPIO_BEMF
41	PA8	TIM1_CH1	
42	PA9	TIM1_CH2	
43	PA10	TIM1_CH3	
46	PA13	SYS_JTMS-SWDIO	TMS
49	PA14	SYS_JTCK-SWCLK	TCK
55	PB3*	SYS_JTDO-TRACESWO	SWO